SHOWCASE

A BAYESIAN BASED AREA TRACKING FOR MONOCULAR VIDEO

YANNIX

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WIL Project

ABSTRACT

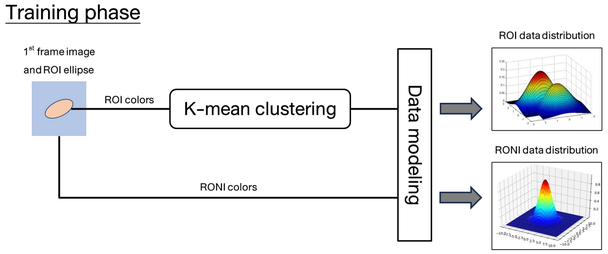
Match moving is a visual effects technique, which is used to determine the camera movement in post-production. This research proposed the technique that can identify and track the area of interest, instead of a point-based feature, in the video sequence. The target area of interest will be selected by an operator in the first frame, then evaluated and tracked by color and homogeneity. In the next frame, the best matched area will be selected using Bayesian approach. The tracking performance can also be improved using the incremental learning technique.

PURPOSE

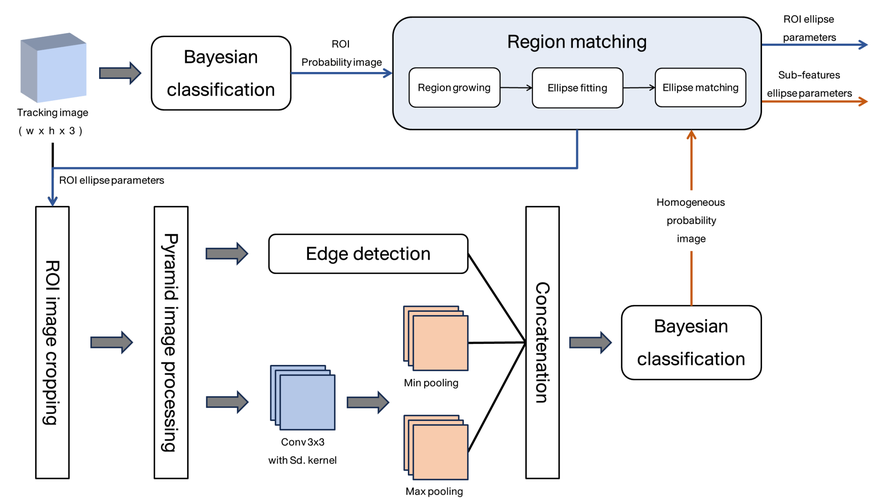
- To study and apply Bayesian learning principles together with the homogeneity and color characteristic to create an area tracking system in video sequence.
- To study the guidelines and limitations of the Bayesian area tracking system in ROI area on video.
- To study the process and outstanding characteristics of homogeneity and color.
- Create a proof-of-concept system for Bayesian area tracking to guide development and application.

METHODOLOGY

Training phase

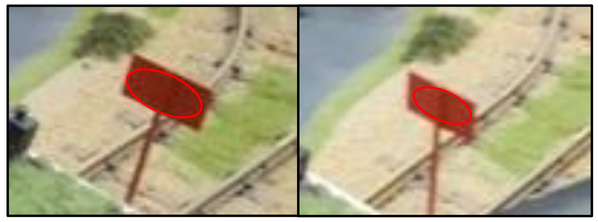


Tracking phase

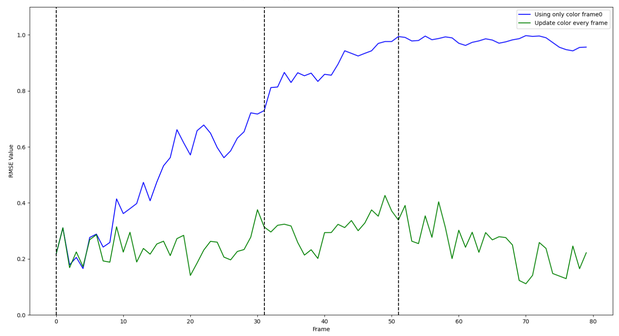


Results

Training image | Tracking image



RMSE probability image compared to ground truth.



The result RMSE of incremental learning system (green line) compared with standard system (blue line)

CONCLUSION

- This method requires only the first image for training.
- The computational time on 1080p images is approximately 5 seconds per frame, which can be reduced by parallel processing.
- The tracking performance is sub-optimal in some scenario due to the simple metrics chosen in this method.